CATwin-IDS: Context-Aware Intrusion Detection System for Both In-Vehicle and External-Vehicle Networks via Digital Twin

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***Abstract*—With the rapid development of the Internet of Vehicles (IoV), the tight coupling between In-Vehicle Networks (IVN) and External Vehicle Networks (EVN) has made vehicular systems vulnerable to sophisticated cross-network attack chains. Existing Intrusion Detection Systems (IDS), however, typically operate in isolation on either IVN or EVN, and lack effective context-aware mechanisms for capturing inter-domain dependencies. To overcome this limitation, we propose CATwin-IDS, a context-aware intrusion detection framework that integrates digital twin technology with a lightweight Distilled Bidirectional Encoder Representations from Transformers (DistilBERT) model. In our design, Conditional Mutual Information (CMI) and Borderline Synthetic Minority Over-sampling Technique (Borderline-SMOTE) are applied for feature optimization and data balancing, while Temporal Self-Attention (TSA) enhances the modeling of spatiotemporal dependencies across heterogeneous traffic. The digital twin provides real-time bidirectional synchronization and a simulation environment, enabling proactive adaptation to dynamic threats. Experimental results on benchmark datasets (Car-Hacking, CICIoV2024, CICIDS2018, CICIoT2023) demonstrate that CATwin-IDS achieves higher accuracy and real-time efficiency compared with state-of-the-art methods, providing a holistic solution for securing IoV against cross-network intrusions.**

***Index Terms*—IoV Security, Intrusion Detection System, DistilBERT, Digital Twin, Pre-trained Transformer Model**

I. Introduction

THE rapid advancement of the Internet of Vehicles (IoV) has revolutionized the automotive industry. As vehicles become increasingly connected, they communicate with diverse entities, enhancing convenience and operational efficiency [1]. However, this connectivity also exposes the automotive ecosystem to numerous security risks [2]. The IoV framework consists of the In-Vehicle Network (IVN) [3] and the External Vehicle Network (EVN) [4]. The IVN coordinates communication among Electronic Control Units (ECUs), primarily employing Controller Area Network (CAN) bus technology, which inherently lacks authentication mechanisms [5]. This weakness allows attackers to infiltrate the IVN via physical interfaces such as Universal Serial Bus (USB) or On-Board Diagnostics (OBD) [6]. In contrast, the EVN leverages Vehicle to Everything (V2X) technology for external interactions and faces wireless threats through Wireless Fidelity (Wi-Fi), Bluetooth, or cellular networks [7], as well as physical-layer vulnerabilities in emerging ambient backscatter NOMA systems [8] [9]. Moreover, the IoV relies on data exchange with cloud services via Road Side Units (RSUs) [10], which introduces additional vulnerabilities.

The IVN and EVN are deeply interdependent through vehicle gateways and cloud platforms [11]. EVN-acquired data (e.g., road conditions, traffic signals) directly influences IVN control decisions (e.g., speed modulation, route planning), while IVN status data (e.g., vehicle speed, diagnostic information) is transmitted externally. This bidirectional flow enables new attack vectors: adversaries may exploit EVN vulnerabilities to compromise the IVN or leverage IVN flaws to disrupt EVN communications. Consequently, siloed Intrusion Detection Systems (IDS) focusing on IVN or EVN alone fail to address these cross-network attack chains. A context-aware intrusion detection approach that jointly analyzes heterogeneous data sources is therefore essential.

Digital twin technology offers unique opportunities for enhancing IoV security. By creating real-time digital replicas of vehicles and networks, digital twins enable bidirectional synchronization and dynamic context modeling between physical and virtual environments [12]. Unlike static IDS frameworks, a digital twin-based IDS can simulate cross-network attacks, generate synthetic rare samples, and adapt detection models to evolving conditions. Particularly in the IoV domain, its ability to integrate multi-modal sensor inputs (Radio Detection And Ranging (RADAR), Light Detection and Ranging (LiDAR), Global Positioning System (GPS), cameras) [13] with network data enables comprehensive situational awareness. Thus, digital twin technology not only provides a simulation environment but also acts as a context provide for intrusion detection.

The increasing sophistication of cyberattacks further underlines the urgency of this approach. For example, a German security expert demonstrated a Tesla vulnerability in 2022 that allowed remote exploitation of 25 vehicles across 13 countries by leveraging EVN flaws to gain IVN control [14].

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Traditional IDS methods-whether anomaly-based, rule-based, or signature-based [15]-struggle to adapt to such multi-domain, evolving threats. Although machine learning methods have been applied to CAN traffic and V2X data, they are limited by heterogeneous encoding formats and protocol diversity [16]. More importantly, most IDS operate in isolation, ignoring real-time IVN–EVN interactions, and therefore lack context-awareness in detecting coordinated attacks. In this work, "context" refers to the mutual dependencies among traffic features, such as temporal sequences, cross-protocol interactions, and statistical co-variations, within a shared operational window. Inspired by BERT-style contextual representation learning, each feature (e.g., CAN ID, packet size, or V2X field) is treated as a token whose semantic meaning is dynamically shaped by its surrounding features and the real-time vehicle state provided by the digital twin.

Recent advances in pre-trained Transformer models [17] inspire new opportunities. Their sequence modeling capabilities align well with vehicular data streams, where CAN frames and V2X payloads exhibit complex temporal dependencies. To balance effectiveness and efficiency, we adopt Distilled Bidirectional Encoder Representations from Transformers (DistilBERT) with only three Transformer layers, and enhance it with a Temporal Self-Attention (TSA) module. This design captures temporal dependencies in traffic data while meeting the stringent latency and resource constraints of in-vehicle devices.

Based on these motivations, we propose CATwin-IDS: a Context-Aware Intrusion Detection System for both IVN and EVN via digital twin. CATwin-IDS integrates a lightweight DistilBERT model with temporal attention and leverages digital twin technology for real-time context fusion, attack simulation, and adaptive detection. This synergy enables efficient modeling of cross-network contextual dependencies while ensuring practical deployability in edge environments.

The main contributions of this paper are summarized as follows:

- We propose CATwin-IDS, a context-aware intrusion detection system empowered by digital twin technology. It fuses IVN and EVN features in real time, enabling cross-domain context modeling and adaptive detection of coordinated attacks.
- We design a hybrid feature engineering approach integrating Conditional Mutual Information (CMI) and Borderline Synthetic Minority Over-sampling Technique (Borderline-SMOTE), improving feature relevance while addressing severe class imbalance in heterogeneous vehicular traffic.
- We develop a lightweight DistilBERT model with TSA, which captures both semantic-rich EVN payloads and time-critical IVN dependencies, thus modeling cross-network contextual correlations under in-vehicle resource constraints.
- We construct a hierarchical digital twin architecture (physical, digital twin, application) that supports bidirectional synchronization, provides a simulation environment for cross-network attack scenarios. This integration enhances the IDS with real-time adaptability and context-awareness.

The remainder of this paper is organized as follows. Section II discusses the related work. Section III presents the proposed model and its key technologies. Section IV describes the experimental results. Section V concludes the paper.

II. Related Work

To comprehensively address the challenges of intrusion detection in IoV systems, this section reviews existing research from three perspectives: detection methods for IVN intrusions, strategies to tackle EVN intrusion risks, and hybrid frameworks integrating IVN and EVN detection. These existing studies on single-network and hybrid network approaches are summarized in Table I.

A. In-Vehicle Network Intrusion Detection

In the development of connected and autonomous vehicles, securing IVNs through robust intrusion detection has become increasingly critical. Within the complex ecosystem of IVNs, intrusion detection technologies have overwhelmingly centered on protecting the CAN bus-primarily because it serves as the backbone for data communication between in-vehicle ECUs. As the primary medium for transmitting in-vehicle data, CAN messages and packets are relayed through this bus architecture. Specifically, a CAN data packet, as depicted in Fig. 1, consists of seven key fields: frame start, arbitration field, control field, data field, Cyclic Redundancy Check (CRC) Field, Acknowledgment (ACK) Field, and frame end [29]. Among these, the Data Field is the most vulnerable to exploitation by attackers, as it contains the actual transmitted data that determines the operations of the ECUs, such as throttle control, brake signals, and steering angle adjustments. Attackers can infiltrate or control the vehicle by injecting malicious messages into the Data Field of the CAN data packet [30].

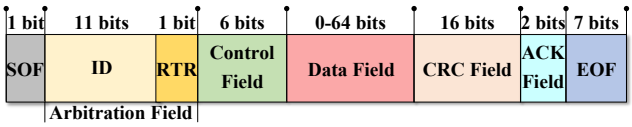


Fig. 1: CAN packet structure.

In recent years, many researchers have focused on IDS based on the CAN bus. In [19], a binary classification detection system based on a hybrid Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN) model was proposed. Ref. [21] and others used CNN deep learning models to experiment with real vehicle data, focusing on specific types of attacks. The authors of [31] applied a multi-stage method combining Artificial Neural Network (ANN) and LSTM, achieving an F1 score of 0.95 and a detection rate of 99.99% on benchmark datasets published by [32] and others. In [33], a scheme was proposed to detect intrusions by establishing voltage fingerprints for each ECU's ID, with experiments conducted on three real vehicles.The study [34] introduced a cluster-based open set recognition method for

TABLE I: Comparison of Related work

Work	Date of Publication	Scenarios	Dataset	Classification Tasks	Main Method	Data Representation	Advantages	Limitations
Yang *et al.* [18]	Jan., 2022	Both	Car-Hacking, CICIDS2017	Multi-class	DT, RF, ET, XGBoost	Feature design	High detection rate, low computational cost	Parameter tuning challenges
Agrawal *et al.* [19]	Nov., 2022	Intra-vehicle	CAN-Intrusion	Binary	LSTM, CNN	Feature design	Detects zero-day attacks, simple architecture	Can not handle multiclass issue
Wang *et al.* [12]	Nov., 2023	Inter-vehicle	UNSW-NB15, CICIDS2017	Multi-class	BiLSTM, CNN, Digital Twin	Feature design	Model fusion spatio-temporal features	Neglects rare attack detection
Anbalagan *et al.* [11]	Dec., 2023	Both	Collected	Multi-class	CNN	Transform into image	High accuracy	Dataset source problem
Ben Said *et al.* [20]	Dec., 2023	Inter-vehicle	NSL-KDD, UNSW-NB15	Binary, Multi-class	BiLSTM, CNN	Feature design	Relatively high detection accuracy	Time-consuming
Wang *et al.* [21]	Jan., 2024	Intra-vehicle	Real-world	Multi-class	CNN	Confederate attack on sensor data fusion	Detection of real datasets	Focus on a specific type of attack
Lai *et al.* [22]	Feb., 2024	Inter-vehicle	UNSW-NB15, CICIDS2018	Multi-class	Transformer, Differential privacy	Transform into text	Improved detection rate of low-probability attacks and privacy protection	Computationally demanding
Jeong *et al.* [23]	Mar., 2024	Intra-vehicle	Collected	Multi-class	BiLSTM	Data bytes of CAN messages	Detects zero-day attacks, high accuracy and interpretability	Hardware dependent
Fu *et al.* [1]	May, 2024	Both	CICIDS, BoT-IoT	Multi-class	BERT	Transform into text	High accuracy, contextual understanding	Neglects rare attacks, computationally demanding
Chen *et al.* [24]	Aug., 2024	Both	UNSW-NB15, CAN-Intrusion, CICIDS2017	Binary	Bi-LSTM, Differential privacy	Feature design	Detects zero-day attacks, protects privacy, high F1	Can not handle multiclass issue
Cui *et al.* [25]	Dec., 2024	Inter-vehicle	UNSW-NB15	Multi-class	OCSVM, iForest	Feature design	Detects zero-day attacks, protects privacy, lightweight	Detection performance of each attack type was not evaluated
Gao *et al.* [26]	Dec., 2024	Both	Car-Hacking, CICIoT2023	Multi-class	BiLSTM, CNN	Transform into text sequence	High accuracy	Real-time detection needs improvement
Li *et al.* [27]	Dec., 2024	Both	Real-world	Binary	BERT	Transform into text sequence	Detection of real datasets	BERT model training time is long
Ullah *et al.* [28]	Mar., 2025	Inter-vehicle	CICIDS2017, UNSW-NB15	Binary, Multi-class	XGBoost, RF, DT, Gradient Boosting	Feature design	Applies Stacking ensemble learning with high accuracy	Lacking validation with recent algorithms

CAN bus intrusion detection, which can detect known categories and identify unknown attacks. It achieved accuracies of 99.37%, 99.36%, 98.93%, and 98.09% on the Car-Hacking dataset [35] for Dos, Fuzzy, Gear, and Rpm attack types, respectively. Although the abovementioned machine-learning-based IDS are efficient and effective, they often lack interpretability. The work [36] addresses this by integrating a threshold-based IDS with a machine learning classifier for attack type categorization, providing explanations for the IDS. Li *et al.* [27] proposed a cloud-collaborative intrusion detection and prevention system that integrates the Car-Hacking dataset, survival analysis dataset, and data collected from real or simulated vehicular environments. By utilizing the contextual modeling capabilities of the Bidirectional Encoder Representations from Transformers (BERT) model, the system classifies in-vehicle traffic. Additionally, they introduced the Encode 2 ID algorithm to encode malicious traffic, generating a unique ID for each traffic instance, thereby saving storage space.

In summary, research on IVN intrusion detection has not only promoted an in-depth understanding of CAN bus attack patterns but also offered important references for protocol feature mining and anomaly modeling in IVN. However, existing methods, while effective for single-bus protocol defense, insufficiently address the dynamic interplay between IVN and external networks-lacking multi-protocol compatibility, cross-

scenario generalization, and interpretability, which limits their applicability to the coupled nature of IoV.

B. External Network Intrusion Detection

Network intrusion detection in the EVN within the IoV has also attracted widespread attention from researchers. In [12], an IoT-based attack behavior recognition system was proposed, which integrates spatiotemporal feature fusion. This system uses a simplified CNN model to learn the spatial features of attacks and a Bidirectional Long Short-Term Memory (BiLSTM) model to capture the temporal features of attacks. Experiments were conducted on the UNSW-NB15 [37] and CICIDS2017 datasets [38]. To address the issue of detecting low-probability attacks, [22] proposed an intrusion detection system based on Transformer models. In [25], machine learning techniques such as One-Class Support Vector Machine (OCSVM) and isolation Forest (iForest) were employed to detect both known and zero-day attacks. [39] introduced an IDS based on Deep Transfer Learning (DTL) that employs a three-tier architecture integrating CNN, Genetic Algorithms (GA), and bootstrap aggregation techniques. By transforming cybersecurity datasets into image data and optimizing model hyperparameters using genetic algorithms, this system achieved a 100% accuracy rate on the Edge-IIoTset dataset [40], which includes 14 types of attacks.

Research on EVN intrusion detection has made progress in spatio-temporal feature fusion, transfer learning, Transformer-based models, and the application of large language models like BERT in text-based data processing. Nevertheless, existing methods are mostly designed for independent EVN scenarios with high system model complexity, and their adaptability in resource-constrained in-vehicle edge environments still needs to be strengthened. This deficiency further heightens the urgent demand for the realization of collaborative defense between in-vehicle and external networks to address the practical challenges in EVN security.

C. Hybrid Intrusion Detection

Recent studies have explored hybrid IDS for integrated IVN-EVN security, aiming to address cross-network attack scenarios. In [18], an ensemble method combining Decision Trees (DT), Random Forest (RF), Extremely Randomized Trees (ET), and XGBoost was proposed, achieving accuracies of 99.99% and 99.88% on the Car-Hacking (representing IVN) and CICIDS2017 (representing EVN) datasets, respectively. In [11], a CNN was used to convert network numerical data into images, making it easier to identify attack types, and the method was validated on the collected dataset. In [1], the relationship between network traffic was considered, and data was converted into text based on a BERT model. The system was validated on the CICIDS and BoT-IoT [41] datasets. In [24], a BiLSTM-based intrusion detection system was employed for binary classification, and Federated Learning was utilized to protect data privacy. Performance validation was conducted on the UNSW-NB15, CAN-Intrusion [32], and CICIDS2017 datasets. Hybrid frameworks have explored the collaborative defense ideas between IVN and EVN, showing potential in improving the coverage of cross-network attack detection. However, existing methods are either limited by complex architectures resulting in insufficient real-time performance or have limitations in the depth of heterogeneous data fusion and the refinement of contextual modeling, making it difficult to fully capture the dynamic coupling relationship between in-vehicle and external networks.

Existing intrusion detection research has explored cross-network hybrid architectures for the IoV, but these frameworks exhibit several limitations. Hybrid architectures achieve cross-network detection through approaches like multi-model integration but typically retain bulky structures, making them unsuitable for resource-constrained in-vehicle edge environments. Most importantly, they fail to fully capture the time-critical CAN frame dependencies between the IVN and EVN, with shortcomings in multi-classification capability, detection performance, and interpretability further restricting their practicality.

Recent studies have begun to explore efficient adaptations of pre-trained Transformer models to address these challenges. For instance, Li *et al.* [42] proposed a knowledge distillation framework that transfers semantic knowledge from BERT to a lightweight CNN-BiLSTM model, achieving over 98% accuracy while significantly reducing computational overhead. Bimmo *et al.* [43] introduced Fed-CALiBER, a federated learning system integrating a lightweight BERT variant for real-time (< 4 ms/sample) CAN bus intrusion detection on edge devices, demonstrating the feasibility of privacy-preserving LLM deployment in distributed vehicular settings. Meanwhile, Sharma *et al.* [44] leveraged pre-trained Transformer models for context-aware authentication in V2V/V2I communications, highlighting the potential of pre-trained Transformer models in building adaptive, intelligent security mechanisms beyond mere detection. These advances collectively signal a shift toward lightweight, context-sensitive, and privacy-aware pre-trained Transformer model applications in automotive cybersecurity-trends that directly motivate our architectural choices.

Notably, EVN-focused research [1], which leverages BERT's sequence modeling capabilities, has inspired our work by demonstrating the potential of pre-trained Transformer models to capture contextual connections in vehicular network data. Building upon this insight, we adopt DistilBERT-a 6-layer lightweight variant of BERT obtained via knowledge distillation-as the foundation of our framework. To further adapt it to the stringent real-time requirements of IoV, we streamline the encoder depth from six to three layers and refine the attention mechanism into TSA. This twofold optimization not only alleviates the computational burden compared to both BERT and standard DistilBERT, but also enhances the model's ability to capture temporal dynamics in cross-network traffic. Combined with digital twin-driven dynamic adaptation, the framework effectively addresses the limitations of existing single-network and hybrid-network IDS approaches.

III. System Model

In this section, we introduce the proposed CATwin-IDS system model in detail, as illustrated in Fig. 2. First, we outline

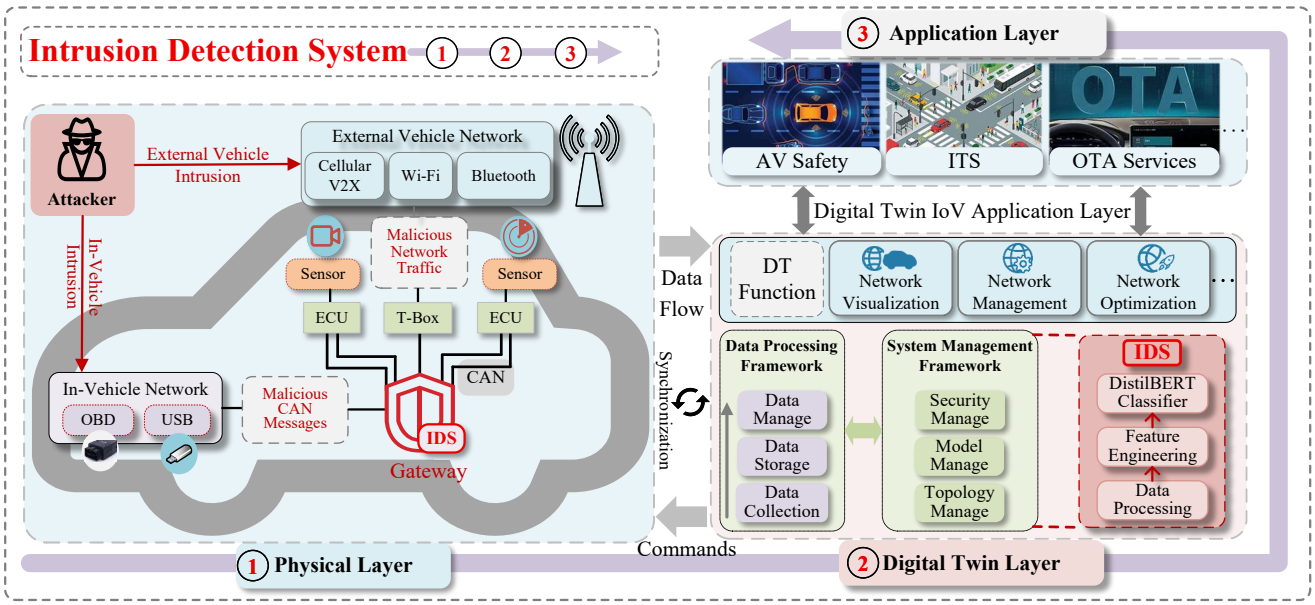


Fig. 2: Application scenario diagram of digital twin intrusion detection system for IoV.

the overall IoV digital twin architecture, which is organized into three layers-physical, digital twin, and application-to enable bidirectional synchronization and cross-domain feature fusion. Next, we describe the training framework of the proposed IDS, consisting of data preprocessing, feature engineering, and the Lightweight DistilBERT Classifier, supported by digital twin simulation. The following subsections elaborate on the preprocessing strategies for heterogeneous vehicular traffic, the hybrid feature engineering methods that enhance discriminative power, and the lightweight DistilBERT model specifically optimized with TSA for resource-constrained environments. Together, these components constitute a cohesive framework that enables efficient and accurate intrusion detection across both IVN and EVN.

A. IoV Digital Twin Architecture Overview

The three-layer architecture of CATwin-IDS is designed to address the dynamic coupling between IVN and EVN, and to bridge the gap between real-world vehicular data acquisition and virtualized intrusion detection. This architecture enables bidirectional data synchronization, cross-domain feature fusion, and secure simulation of attack scenarios-key capabilities for overcoming the limitations of traditional isolated-network IDS. Each layer undertakes distinct but interdependent roles, forming a closed-loop system that supports real-time, adaptive intrusion detection for IoV.

1) Physical Layer: The core communication protocol of IVN is the CAN bus protocol, which allows various Electronic Control Units (ECUs) in the vehicle to receive signals from sensors and make control decisions based on these signals. These ECUs communicate with each other and exchange data with the vehicle gateway via the CAN bus protocol. The vehicle gateway serves as a central hub, not only transmitting data within the vehicle's internal network but also enhancing network security through an integrated IDS, effectively defending against potential network attack threats.

Vehicle interfaces such as USB, OBD, and Unified Diagnostic Services (UDS) provide means for interaction with the external environment but may also introduce security risks related to in-vehicle intrusion. Therefore, these interfaces require appropriate security measures to protect them from potential attacks by external intruders. The Telematics-Box (T-Box), as the vehicle's remote communication unit, provides functions like cellular networks and short-range communication (such as Near Field Communication (NFC), Bluetooth, and Wi-Fi). While these communication capabilities enhance the vehicle's connectivity, they also increase the risk of external vehicle intrusion.

In addition, the physical layer includes all physical devices and sensors, such as GPS positioning systems, high-definition cameras, and LiDAR. These devices are responsible for collecting real-time data on the vehicle, covering key information such as the vehicle's operational status, environmental conditions, and driving behavior. The primary task of the physical layer is to ensure the accuracy and integrity of this data, providing reliable input for the digital twin layer. To achieve this, the physical layer not only needs to efficiently collect and transmit data but also must implement multiple security measures to prevent data from being tampered with or lost, thereby providing a solid foundation for intelligent decision-making and control.

2) Digital Twin Layer: The digital twin layer is responsible for constructing and maintaining a high-fidelity virtual replica of the IoV environment, faithfully reflecting both the IVN and the EVN. Its architecture follows the modular design shown in Fig. 2, consisting of a data processing framework and a system management framework, with the two frameworks interacting dynamically to achieve precise mapping and intelligent man-

agement of IVN-EVN interactions.

The data processing framework integrates continuous real-time streams from physical sensors, ECUs, and external communication modules (e.g., V2X, Wi-Fi). This data is first normalized and stored, ensuring integrity and temporal alignment across domains. The virtual twin's state $\mathbf{s}_t^{\text{twin}}$ is updated through a bidirectional synchronization process:

$$\mathbf{s}_t^{\text{twin}} = (1-\alpha)\,\mathbf{s}_{t-1}^{\text{twin}} + \alpha\,\mathbf{s}_t^{\text{phys}}, \tag{1}$$

where $\alpha \in [0,1]$ controls the update rate between physical measurements $\mathbf{s}_t^{\text{phys}}$ and the previous virtual state $\mathbf{s}_{t-1}^{\text{twin}}$. This mechanism enables the twin to remain consistent with the real system while preserving short-term predictive dynamics, ensuring timely tracking of sudden changes in IVN (e.g., abnormal CAN bus frequency) and EVN (e.g., V2X signal spoofing) for rapid threat identification.

The system management framework incorporates:

- Security Management: Interfaces with the IDS to detect and mitigate threats, validating countermeasures in the virtual space before deployment.
- Model Management: Continuously refines virtual model parameters by comparing simulated and observed outputs.
- Topology Management: Dynamically maps IVN-EVN interconnections, tracking gateway, T-Box, and access point configurations.

The data processing framework provides real-time updated state data as input for the system management framework, while the topology management results from the system management framework guide the priority setting of data streams in the data processing framework, forming a closed-loop optimization mechanism.

Cross-domain feature fusion module aligns heterogeneous IVN and EVN features into a unified embedding space:

$$\mathbf{z} \in \mathbb{R}^d = f_{\text{IVN}}(\mathbf{x}_{\text{IVN}}) \,\|\, f_{\text{EVN}}(\mathbf{x}_{\text{EVN}}), \tag{2}$$

where $\|$ denotes vector concatenation, and $f_{\text{IVN}}, f_{\text{EVN}}$ are domain-specific encoders mapping IVN and EVN features into the joint embedding space $\mathbb{R}^d$. This representation feeds into the Borderline-SMOTE in Sec. III-B, ensuring synthetic samples preserve real-world cross-network dependencies, and directly serves as the input for the Lightweight DistilBERT classifier to support its modeling of spatiotemporal correlations across heterogeneous networks.

During inference, the fused features are continuously streamed into the DistilBERT classifier, enabling real-time and context-aware intrusion detection. In this way, the digital twin not only maintains a synchronized virtual replica of the physical system but also functions as a live feature provider for the detection model. Beyond online detection, the twin further offers a secure experimentation environment in which simulated attack scenarios (e.g., CAN flooding and V2X spoofing) can be executed to support adaptive classifier tuning without risking the physical vehicle.

3) Application Layer: The application layer serves as the interface for end-users, translating the analysis results and detection decisions from the digital twin layer into actionable steps and feedback. Application scenarios include autonomous vehicle security, intelligent transportation systems (ITS), remote software updates and OTA services, and more.

B. Context-Aware Intrusion Detection Framework

CATwin-IDS employs a DistilBERT encoder to learn contextualized feature representations. Through bidirectional self-attention, every feature embedding incorporates information from all other features within the same detection window, capturing both temporal and statistical correlations. The digital-twin layer enriches this process by injecting environment-specific variables, enabling conditional context modeling under real-time vehicle states.

The training framework integrated into the digital twin layer primarily consists of three stages: data preprocessing, feature engineering, and the Lightweight DistilBERT Classifier, as shown in Fig. 3. The pseudocode for the intrusion detection process, including data preprocessing, feature engineering, digital twin simulation, lightweight DistilBERT feature extraction, and final classification decisions, is described in Alg. 1.

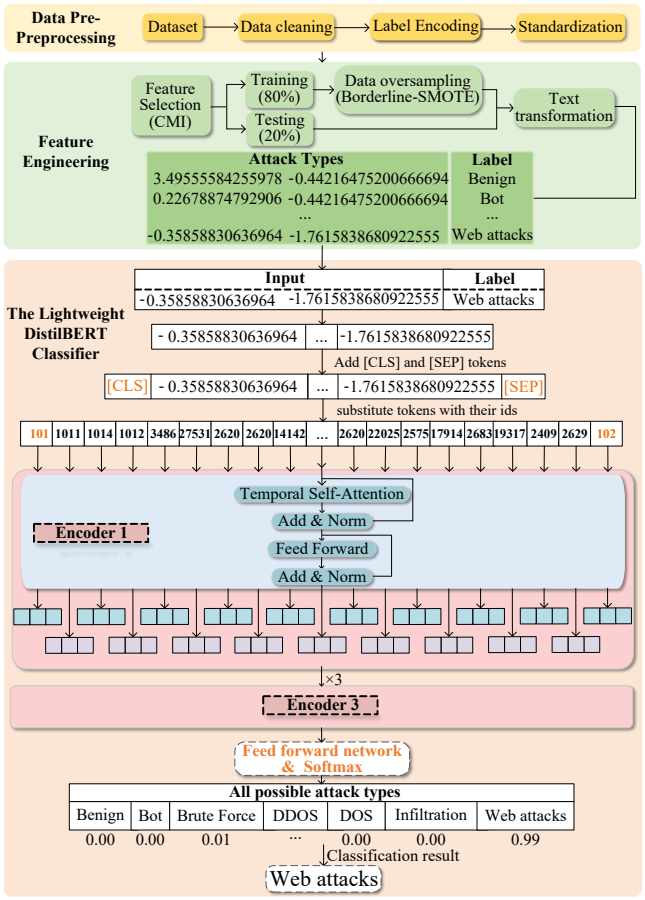


Fig. 3: The overall framework of CATwin-IDS.

1) Data Pre-Processing: In real-world application scenarios, network traffic data typically exhibits class imbalance [45], where the volume of normal traffic significantly exceeds that of anomalous traffic. To enhance the model's ability to detect anomalous traffic and improve its generalization performance

Algorithm 1 Proposed Intrusion Detection Algorithm with Lightweight DistilBERT Classifier

Input: Network traffic data $\mathcal{D}$ (structured data), digital twin environment $\mathcal{T}$;
Output: Predicted labels y;

Step 1: Data Preprocessing
Load dataset $\mathcal{D}$;
Perform data cleaning (e.g., handle missing values and outliers);
Encode categorical labels into numeric values;
Standardize numeric features to ensure consistent scales;

Step 2: Feature Engineering
Select top-k features using *CMI*;
Split $\mathcal{D}$ into training (80%) and testing (20%) subsets;
Apply *Borderline-SMOTE* to balance class distribution in training data;
Transform structured features into textual representations;

Step 3: Digital Twin Simulation
Initialize digital twin environment $\mathcal{T}$ for real-time traffic simulation;

Step 4: Lightweight DistilBERT Feature Extraction
Tokenize textual input: Input_tokens ← Tokenizer(Textual_Representation);
Add `[CLS]` and `[SEP]` tokens;
Pass tokenized inputs through the Lightweight DistilBERT Classifier: All_Layers_Output ← LightweightDistilBERT(Input_tokens);
Extract contextual embeddings from the final (third) encoder layer: Embeddings ← All_Layers_Output.Encoder3;

Step 5: Classification
Feed embeddings into the Lightweight DistilBERT Classifier's output layer (FFNN + softmax);
Compute class probabilities and assign predicted labels: Predicted_Labels ← $\arg\max$(Probabilities);

Step 6: Output
Return the final predicted labels y as intrusion detection outcomes.

in diverse data environments, it is necessary to preprocess the data, transforming raw network traffic data into a format suitable for spatiotemporal feature analysis, thus providing an appropriate data foundation for in-depth data mining and model training.

To address the class imbalance issue, we employed random sampling techniques to balance the sample sizes of different attack types and normal traffic within the dataset. This process ensures the representativeness of each sample type during model training, thereby improving the fairness of the training process and the accuracy of evaluation.

During the data processing, to handle the occurrence of infinite or NaN values, this paper adopted a mean imputation strategy. By replacing infinite or NaN values with the corresponding feature's average value, the integrity and consistency of the data were effectively maintained, providing a reliable foundation for subsequent data analysis. Additionally, to facilitate model processing, LabelEncoder technology was used to encode the labels of the dataset [46]. This technique converts categorical text labels into numerical values, which not only simplifies the model's input requirements but also enhances the efficiency of data processing by the model.

2) Feature Engineering: Building upon data preprocessing, the goal of feature engineering is to further refine and optimize the dataset to improve model training efficiency and classification performance. To address the challenges of modeling cross-domain dependencies between IVN and EVN, traditional single-network feature selection methods can only analyze IVN's CAN bus features (e.g., message ID frequency) or EVN's V2X protocol characteristics, while neglecting the attack chains formed through gateway data synchronization (e.g., spoofed EVN traffic signals infiltrating IVN to disrupt ECU decision-making). To overcome this limitation, this paper integrates multiple feature selection methods-including Mutual Information (MI) for nonlinear correlation analysis, Chi-square test for categorical variable screening, and Principal Component Analysis (PCA) for linear dimensionality reduction-to perform cross-domain feature selection. This approach preserves critical original features while eliminating redundant dimensions, ultimately constructing a low-dimensional yet highly discriminative feature subset through cross-network dependency mining to achieve effective anomalous traffic detection. Below are several commonly used feature selection methods and their mathematical expressions:

- **MI**: MI is a method used to assess the amount of shared information between two random variables. It can capture both linear and nonlinear relationships between variables. The expression for calculating MI is as follows:

$$MI(X,Y) = \sum_{x\in X}\sum_{y\in Y} P(x,y)\log\left(\frac{P(x,y)}{P(x)P(y)}\right), \quad (3)$$

  where X represents cross-network interaction features (e.g., combined IVN-EVN attributes such as CAN ID frequency and V2X packet rate), Y denotes attack categories (e.g., benign, DoS, spoofing), $p(x,y)$ is the joint probability distribution, and $p(x)$ and $p(y)$ are the marginal probability distributions. By computing the MI between each feature and the target variable, followed by MI value ranking, we prioritize retaining features that simultaneously influence both IVN and EVN.
- **CMI**: To further capture the latent cross-domain dependency between IVN and EVN under specific operational contexts (e.g., gateway load state, time window), we extend MI to its conditional form:

$$I(B_k; F \mid Z) = \sum_{z}\sum_{b_k}\sum_{f} p(b_k, f, z)\times \log\frac{p(b_k, f \mid z)}{p(b_k \mid z)p(f \mid z)}, \quad (4)$$

  where B_k denotes the k-th CAN payload byte, F denotes a selected EVN feature (e.g., `Flow_Byts/s`), and Z represents contextual variables such as gateway state or

time interval. We further adopt the normalized form:

$$\mathrm{NCMI}(B_k; F \mid Z) = \frac{I(B_k; F \mid Z)}{\sqrt{H(B_k \mid Z)\, H(F \mid Z)}}, \quad (5)$$

where $H(\cdot \mid Z)$ denotes conditional entropy. This metric allows fair comparison between heterogeneous feature pairs and prioritizes those most indicative of cross-network attack chains.

- **Chi-squared Test**: The chi-squared test is a statistical method used to test the independence between categorical variables. In feature selection, it is used to assess the association between a single feature and the target variable, expressed as:

$$\chi^2 = \sum \frac{(O - E)^2}{E}, \quad (6)$$

where O represents the observed frequency, and E is the expected frequency under the assumption of independence.

- **PCA**: PCA transforms a set of possibly correlated variables into a set of linearly uncorrelated principal components via orthogonal transformation:

$$PCA(X) = XW, \quad (7)$$

where X is the original dataset matrix, and W contains eigenvectors of the covariance matrix.

In this paper, to quantify the interdependence between features and the target variable, we adopt MI and extend it to CMI for cross-domain feature selection, ensuring that selected features capture correlated anomalies in both IVN and EVN. By quantifying these correlations, the system identifies attack propagation patterns across networks (e.g., DoS attacks manifesting as EVN traffic surges alongside excessive CAN bus load in IVN). Following this feature selection, the dataset is partitioned into training (80%) and testing (20%) sets.

To further address the issue of class imbalance, we adopt Borderline-SMOTE, which differs from traditional Synthetic Minority Over-Sampling Technique (SMOTE) by focusing on minority samples located near the decision boundary. Instead of generating synthetic samples uniformly across the feature space, Borderline-SMOTE interpolates new data points in these critical regions where misclassification is most likely to occur. This targeted sampling strategy enhances the model's ability to distinguish between normal and attack traffic, reduces the risk of introducing noisy or redundant samples, and improves classification robustness under imbalanced conditions.

Finally, numeric features are converted into text format for input into the Lightweight DistilBERT Classifier. The text transformation process directly maps structured numeric features to sequential text strings while preserving the original feature order. This design enables the Lightweight DistilBERT classifier to leverage its pre-trained sequence modeling capabilities without requiring additional input layers for heterogeneous numeric features, reducing model complexity. For illustration, a sample from the CICIDS2018 (EVN) dataset, consisting of protocol and traffic-related numeric features, is converted into a text sequence such as "6 64 9306.82 120 0 ...". Similarly, a sample from the Car-Hacking (IVN) dataset, including the CAN ID and eight payload fields, is transformed into a sequence like "1201 41 39 39 35 0 0 0 154". Both IVN and EVN samples follow the same numeric-to-text conversion rule, ensuring unified processing of cross-domain traffic while preserving intrinsic temporal ordering, which further facilitates the TSA mechanism in capturing dynamic traffic patterns.

3) The Lightweight DistilBERT Classifier: The lightweight DistilBERT classifier is a refined variant of the original DistilBERT model, specifically adapted for intrusion detection in the IoV. While standard DistilBERT employs knowledge distillation to compress BERT into a six-layer architecture [47], our variant further reduces the encoder depth to three layers to meet the stringent latency and memory requirements of in-vehicle edge devices. This reduction lowers computational overhead and improves deployability in resource-constrained environments, without sacrificing essential contextual modeling capacity.

Each encoder layer consists of a Temporal Self-Attention (TSA) mechanism, a position-wise feed-forward network (FFN), and residual connections with layer normalization. The TSA mechanism introduces a temporal bias into the attention weights, explicitly capturing the order and intervals of network events:

$$\text{Attention}(Q, K, V, \Delta t) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} - \beta|\Delta t|\right) V, \quad (8)$$

where Q, K, and V denote the query, key, and value, d_k is the key dimension, Δt is the timestamp difference between tokens, and $\beta > 0$ controls the temporal decay. This allows the model to prioritize temporally proximate features, improving detection of time-sensitive anomalies in network traffic.

The FFN performs a nonlinear transformation at each position:

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2, \quad (9)$$

where x is the input, W_1 and b_1 are parameters of the first fully connected layer with ReLU activation, and W_2, b_2 are parameters of the second fully connected layer. Residual connections and layer normalization are applied after both TSA and FFN to ensure training stability:

$$\text{Y} = \text{LayerNorm}(x + \text{Sublayer}(x)), \quad (10)$$

$$\text{LayerNorm}(x) = \gamma\frac{x - \mu}{\sigma} + \beta, \quad (11)$$

where μ and σ are the mean and standard deviation, and γ and β are learnable parameters.

Distinct from the original NLP-oriented DistilBERT that relies on masked token prediction during pretraining, our classifier directly consumes structured vehicular traffic features converted into sequential text form. By treating each numeric feature as an individual token, the Lightweight DistilBERT Classifier effectively transforms heterogeneous vehicular data into a unified representation space. The incorporation of temporal attention further enhances the ability to correlate abnormal EVN communication patterns with time-critical IVN CAN bus signals. As a result, the model achieves robust intrusion detection performance across heterogeneous domains, offering a practical and scalable solution for real-time IoV security.

In summary, the lightweight DistilBERT classifier, combined with TSA, FFN, and residual connections, forms the core of our intrusion detection framework. Integrated with the digital twin layer and preceded by data preprocessing and feature engineering, this architecture enables efficient, context-aware detection of cross-network attacks while meeting the real-time and resource-constrained requirements of IoV environments.

IV. Evaluation Results

This section provides a detailed overview of the experimental design and results analysis, aiming to validate the effectiveness and performance of the proposed CATwin-IDS model in both EVN and IVN. A VMware ESXi virtualization server runs the digital twin of the IoV network environment, replicating the exact topology and functions of the physical IVN and EVN; all traffic data from the physical networks is mirrored to this virtualized server to ensure consistency. First, it introduces the characteristics and sources of the datasets used for training and evaluation. Next, it describes the hardware and software environment, as well as the data preprocessing and sampling methods employed in the experiments. Finally, the chapter comprehensively evaluates the superiority of the CATwin-IDS model across multiple performance metrics through comparisons with various baseline models, conducts an in-depth analysis of its performance using confusion matrices and performance comparison charts, and validates its cross-network intrusion detection efficacy.

A. Experimental Design

This paper involves datasets primarily categorized into two main types: EVN and IVN, aimed at providing the necessary network traffic data for training and evaluating IDS models to determine whether network behaviors are benign. The EVN dataset is sourced from the Canadian Institute for Cybersecurity (CIC), including CICIDS2018 [38] and CICIoT2023 [48], covering various types of network attacks. The CICIoT2023 dataset includes 33 types of attacks, categorized into seven classes: DDoS, DoS, Recon, Web-based, Brute-force, Spoofing, and Mirai attacks, and it contains 39 features. The CICIDS2018 dataset comprises 78 features and multiple types of attacks, including Brute-force, Heartbleed, Botnet, DoS, DDoS, Web application, and Infiltration attacks of the network from inside. The IVN dataset is specifically focused on collecting traffic data related to the CAN bus. This dataset meticulously records feature information, including timestamps, CAN IDs, Data Length Codes (DLC), and the 8-byte data fields (DATA[0]-DATA[7]) in CAN packets, providing essential contextual information for analyzing communications on the CAN bus. The IVN datasets used in this paper for evaluating IDS performance in vehicular network environments include Car-Hacking [35] and CICIoV2024 [49]. By integrating both EVN and IVN datasets, this paper can comprehensively assess the performance of IDS across different network environments, ensuring that the proposed solutions are effective against a diverse range of network attacks and enhancing the overall security of vehicular networks. By integrating both EVN and IVN datasets, this paper can comprehensively assess the performance of IDS across different network environments, ensuring that the proposed solutions are effective against a diverse range of network attacks and enhancing the overall security of vehicular networks.

The experimental setup adopts NVIDIA GeForce RTX 3060 with 12 GB of GPU memory. The experiments were conducted using Python version 3.8.0. Model construction was primarily carried out using PyTorch, version 1.9.0.

The performance evaluation metrics utilized in this paper include accuracy, precision, recall, and F1, derived from four fundamental parameters: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), as provided below.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \tag{12}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \tag{13}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{14}$$

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{15}$$

B. Data Preprocessing

In real-life scenarios, normal data typically constitutes a large proportion, leading to highly imbalanced datasets. This imbalance often undermines the generalization and stability of system models, as they may become biased towards the majority class. To enhance the robustness of the experimental results, we conducted uniform sampling for each type of attack and normal samples, ensuring that the data volume for each type is the same. Due to the large number of attack class labels, we referred to the attack classification standards of the CIC to group class labels with similar characteristics into unified attack types. This classification approach not only aligns with CIC but also enables more effective analysis and evaluation.

This paper considers each type of attack and uniformly collects data for each type as the experimental evaluation object, comparing two datasets each for IVN and EVN. The details of the original datasets, attack types, training sets, and test sets for these four datasets are respectively presented in Table II, Table III, Table IV, and Table V. In the data preprocessing stage, CMI was used to perform feature selection on four datasets. Taking the CICIDS2018 dataset as an example, 18 out of 78 features were chosen as input features. This approach effectively maintains detection capabilities while reducing computational complexity and redundancy. The importance of the selected features from the CICIoT2023 dataset is illustrated in the Fig. 4 below.

TABLE II: Car-Hacking Dataset

Class Label	Original	Train (80%)	Test (20%)
Benign	13432572	16000	4000
DoS	1048576	16000	4000
Fuzzy	1048576	16000	4000
Gear Spoofing	1048576	16000	4000
RPM Spoofing	1048576	16000	4000

TABLE III: CICIoV2024 Dataset

Class Label	Attack Type	Original	Train (80%)	Test (20%)
Benign	Benign	1048575	80000	20000
DoS	DoS	74663	80000	14933
GAS RPM SPEED STEERING-WHEEL	Spoofing	109819	80000	20000

TABLE IV: CICIDS2018 Dataset

Class Label	Attack Type	Original	Train (80%)	Test (20%)
Benign	Benign	13432572	80000	20000
Bot	Bot	286191	80000	20000
DoS attacks-Hulk DoS-SlowHTTPTest DoS-GoldenEye DoS-slowloris	DoS	706798	80000	20000
DDOS-HOIC DDOS-LOIC-UDP	DDOS	687742	80000	20000
Brute Force-Web Brute Force-XSS SQL Injection	Web attacks	566	80000	113
FTP-BruteForce SSH-BruteForce	Brute Force	380948	80000	20000
Infiltration	Infiltration	161934	80000	20000

C. Model Comparison and Performance Analysis

To comprehensively evaluate CATwin-IDS across different methodological paradigms, we select 14 baseline models covering (1) traditional machine learning methods (KNN, DT, ET, SVM, LightGBM, CatBoost, XGBoost) representing tree-based and distance-based classifiers; (2) sequential deep networks (LSTM, BiLSTM, RNN, CNN) focusing on temporal and local feature learning; and (3) Transformer-based models (BERT, Distilling BERT for Natural Language Understanding (TinyBERT), Transformer) for contextual feature understanding and efficiency comparison. The results of these comparisons are shown in Table VI and Table VII. The CATwin-IDS model achieved an F1 score of 100% across all types of attacks on the Car-Hacking and CICIoV2024 datasets. In the more complex EVN datasets, namely CICIDS2018 and CICIoT2023, the CATwin-IDS model adopted in this

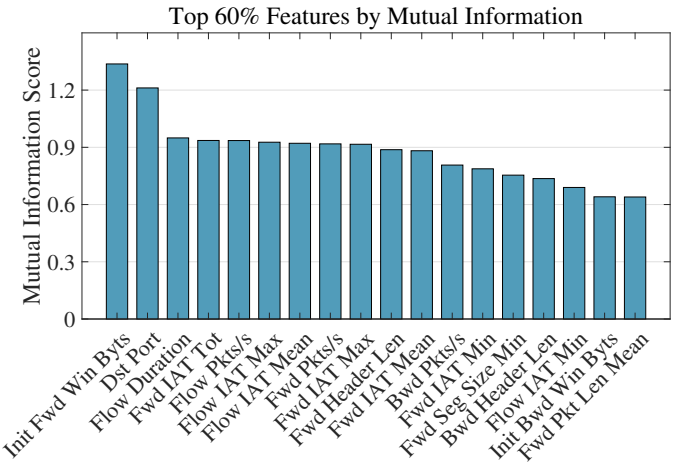


Fig. 4: Selected Feature Importance from CICIDS2018.

TABLE V: CICIoT2023 Dataset

Class Label	Attack Type	Original	Train (80%)	Test (20%)
Benign	Benign	362361	80000	20000
Dictionary brute force	Brute Force	13064	80000	2613
Arp spoofing DNS spoofing	Spoofing	426604	80000	20000
TCP flood HTTP flood SYN flood UDP flood	DoS	730344	80000	20000
ACK fragmentation UDP flood SlowLoris ICMP flood RSTFIN flood PSHACK flood HTTP flood UDP fragmentation TCP flood SYN flood SynonymouslP flood	DDoS	1707878	80000	20000
Ping sweep OS scan Vulnerability scan Port scan Host discovery	Recon	690534	80000	20000
Sql injection Command injection Backdoor malware Uploading attack XSS Browser hijacking	Web-based	24829	80000	4966
GREIP flood Greeth flood UDPPlain	Mirai	105151	80000	20000

TABLE VI: CICIDS2018 Dataset Evaluation Metrics

Model \ Metrics	Accuracy	Precision	Recall	F1
KNN	0.8732	0.8835	0.8732	0.8720
DT	0.8920	0.8926	0.8920	0.8919
ET	0.8969	0.8975	0.8969	0.8968
SVM	0.8724	0.8743	0.8724	0.8716
LightGBM	0.8700	0.8731	0.8700	0.8674
CatBoost	0.9075	0.9098	0.9075	0.9072
XGBoost	0.9076	0.9104	0.9076	0.9072
BERT	0.9091	0.9110	0.9091	0.9089
TinyBERT	0.9040	0.9072	0.9040	0.9034
CNN	0.8748	0.8904	0.8748	0.8702
LSTM	0.8735	0.8768	0.8735	0.8724
BiLSTM	0.8782	0.8867	0.8782	0.8759
RNN	0.8351	0.8390	0.8351	0.8337
Transformer	**0.8856**	**0.8879**	**0.8856**	**0.8845**
CATwin-IDS	**0.9107**	**0.9129**	**0.9107**	**0.9104**

paper demonstrated outstanding performance, leading other models with accuracy scores of 0.9097 and 0.8396, and F1 scores of 0.9094 and 0.8382, respectively. The comparison includes a standard Transformer model, which achieved accuracy of 0.8856 and F1 of 0.8845 on CICIDS2018, and accuracy of 0.8277 and F1 of 0.8250 on CICIoT2023. When compared with traditional machine learning models, CATwin-IDS showed higher performance on both the CICIDS2018 and CICIoT2023 datasets. Although CatBoost and XGBoost performed well on the CICIDS2018 dataset, and LightGBM, CatBoost, and XGBoost also showed good performance on the CICIoT2023 dataset, their accuracy and F1 scores were slightly lower than those of DistilBERT. This indicates po-

TABLE VII: CICIoT2023 Dataset Evaluation Metrics

Model \ Metrics	Accuracy	Precision	Recall	F1
KNN	0.7195	0.7162	0.7195	0.7142
DT	0.7861	0.7862	0.7861	0.7861
ET	0.8164	0.8154	0.8164	0.8133
SVM	0.7723	0.7834	0.7723	0.7626
LightGBM	0.8359	0.8373	0.8359	0.8331
CatBoost	0.8378	0.8387	0.8378	0.8347
XGBoost	0.8356	0.8366	0.8356	0.8324
BERT	0.8370	0.8359	0.8370	0.8348
TinyBERT	0.8318	0.8332	0.8318	0.8276
CNN	0.8014	0.8094	0.8014	0.8000
LSTM	0.7917	0.7991	0.7917	0.7829
BiLSTM	0.8132	0.8105	0.8132	0.8039
RNN	0.7607	0.7618	0.7607	0.7583
Transformer	**0.8277**	**0.8285**	**0.8277**	**0.8250**
CATwin-IDS	**0.8423**	**0.8432**	**0.8422**	**0.8414**

tential limitations of these models in handling complex data. Furthermore, DistilBERT maintained performance similar to BERT while reducing model complexity and computational resource requirements, thereby enhancing efficiency in practical applications. Compared to TinyBERT, another lightweight BERT model, DistilBERT demonstrated advantages in both resource consumption and performance metrics. These results indicate that DistilBERT not only provides high accuracy in classification tasks but also maintains high recall and precision.

An ablation experiment was conducted to investigate the impact of encoder depth on long-range temporal dependency capture and model efficiency. As shown in Table VIII, the 3-layer configuration, integrated with the TSA mechanism, achieves an optimal balance between long-range dependency modeling capability and computational efficiency. Specifically, this design reduces relative computational cost by 49.2% and improves inference efficiency by 46.4% compared to deeper architectures, while attaining an F1-score of 0.9104 that outperforms the 2-layer TSA-integrated variant by 0.25%. In contrast, the 2-layer configuration exhibits insufficient capacity to capture complex long-range dependencies, and the 4-layer configuration delivers only marginal performance changes while incurring a 41.4% increase in convergence time. This validates our design choice of the 3-layer structure for lightweight and real-time network intrusion detection scenarios.

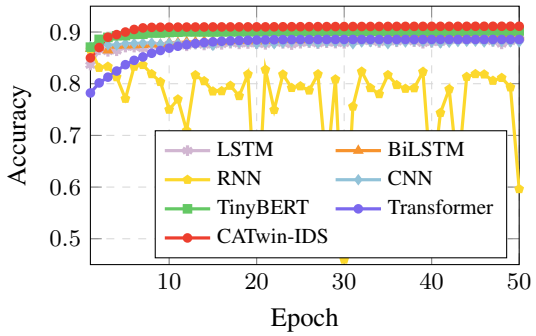
TABLE VIII: Ablation Experiment Results on CICIDS2018 Dataset

Model Configuration	F1 Score	Convergence Time (s)	Stability
2-layer DistilBERT + TSA	0.9079	34680.78	±0.0002
3-layer DistilBERT + TSA	**0.9101**	**32551.43**	**±0.0013**
4-layer DistilBERT + TSA	0.9103	45992.71	±0.0010
6-layer DistilBERT + TSA	0.9099	63902.37	±0.0003

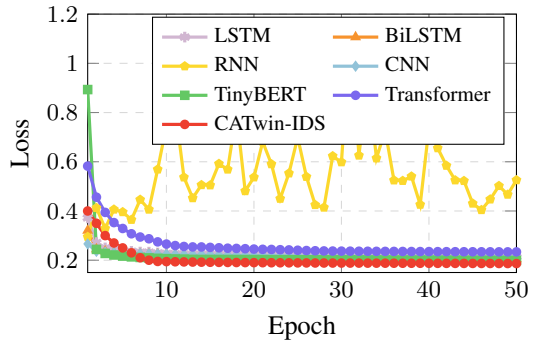
D. Classification Performance and Confusion Matrix Analysis

The proposed CATwin-IDS framework was systematically evaluated against LSTM, BiLSTM, RNN, CNN, TinyBERT and Transformer models across multiple datasets. For CATwin-IDS, the hyperparameters were configured as follows: a base layer dropout rate of 0.25, a sequence classification layer dropout rate of 0.3, the AdamW optimizer (learning rate of 8×10^{-6}, weight decay coefficient of 0.1, bias correction disabled), a batch size of 32, and a fixed training duration of 50 epochs. To ensure fair comparison, identical preprocessing pipelines and training epochs were applied to all models. TinyBERT was trained with the same hyperparameter configuration as CATwin-IDS to validate transferability under equivalent resource constraints, while LSTM, BiLSTM, RNN, and CNN followed widely adopted baseline configurations in the IDS literature:

- **LSTM/BiLSTM/RNN**: 50 hidden units → 64-dimensional fully connected layer (ReLU activation) → Dropout(0.2) → classification layer.
- **CNN**: 64 convolutional kernels (kernel size 3) → max pooling → flattening → 64-dimensional fully connected layer (ReLU activation) → Dropout(0.2) → classification layer.



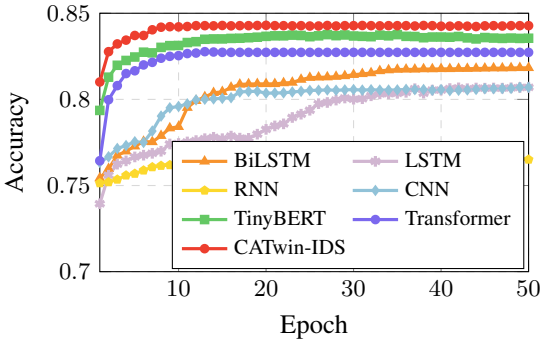
(a) Test accuracy on CICIDS2018 dataset



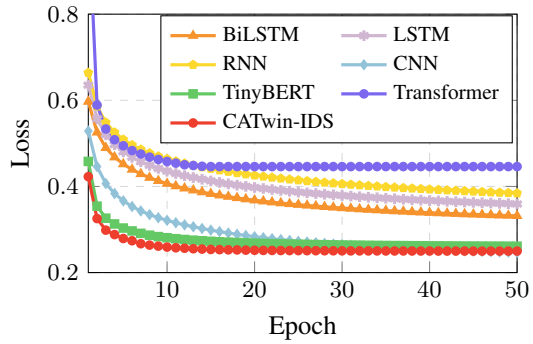
(b) Train loss on CICIDS2018 dataset

Fig. 5: Comparison of EVN performance indicators on CICIDS2018 dataset.

On CICIDS2018 (Fig. 5(a), 5(b)), CATwin-IDS demonstrates faster convergence (loss stabilizing below 0.20 after 20 epochs) and higher test accuracy than LSTM, BiLSTM, CNN, and RNN, the latter showing substantial instability with fluctuating losses. On CICIoT2023 (Fig. 6(a), 6(b)), CATwin-IDS achieves accuracy gains of 1.1% over TinyBERT while converging earlier (15 epochs). Under identical hardware (NVIDIA GeForce RTX 3060 GPU), CATwin-IDS requires 252 seconds per validation epoch, compared to 871 seconds for BERT, corresponding to a 71.1% reduction in runtime. This relative efficiency gain demonstrates the effectiveness of our streamlined architecture (e.g., 3-layer Transformer, TSA) in balancing accuracy and computational cost. On Car-Hacking and CICIoV2024 (Figs. 7 and 8), CATwin-IDS consistently

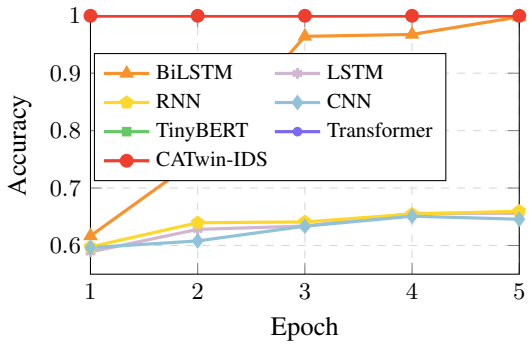


(a) Test accuracy on CICIoT2023 dataset

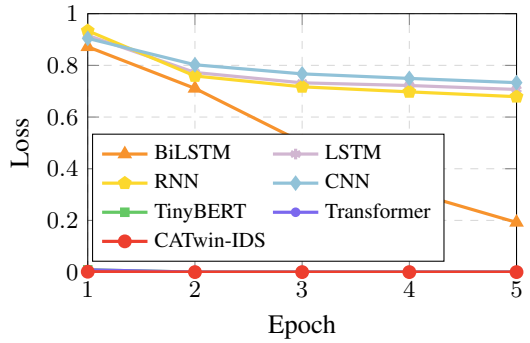


(b) Train loss on CICIoT2023 dataset

Fig. 6: Comparison of EVN performance indicators on CICIoT2023 dataset.



(a) Test accuracy on Car-Hacking dataset

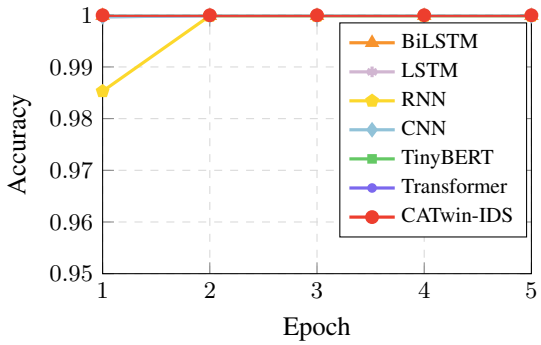


(b) Train loss on Car-Hacking dataset

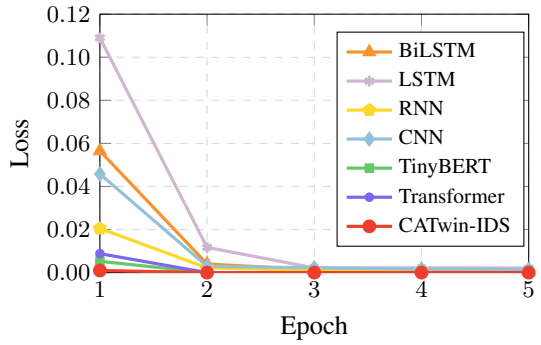
Fig. 7: Comparison of IVN performance indicators on Car-Hacking dataset.

achieves accuracy above 0.99 with loss below 0.02, validating its robustness and generalizability across heterogeneous vehicular domains for reliable cross-network intrusion detection. While in-vehicle edge devices (with stricter resource constraints than GPUs) demand even lower latency, this result highlights a critical step toward real-time responsiveness, outperforming heavyweight models like BERT in resource efficiency, which is foundational for deployment on automotive-grade hardware. Future work will further optimize via model quantization and hardware-aware pruning to align with the stringent real-time requirements of IoV.

Fig. 9 presents the confusion matrices of the classification performance of the proposed system across four distinct datasets, which span both IVN and EVN scenarios, including Car-Hacking, CICIoV2024, CICIDS2018, and CICIoT2023. The results indicate that the system generally performs well in identifying the BENIGN (normal traffic) category but shows some confusion in differentiating certain types of attacks, such as DoS and Bot. Particularly in the CICIDS2018 dataset, the system exhibits high accuracy in recognizing DoS and Bot attacks, while in the CICIoT2023 dataset, the performance is also commendable for Bot and DoS categories. However, for complex attack patterns like Web attacks and Infiltration, there is room for improvement in the system's performance. These findings suggest that although the proposed system demonstrates robust classification capabilities across multiple datasets, further refinement is needed to enhance its detection accuracy and robustness, especially for specific types of cyber attacks.

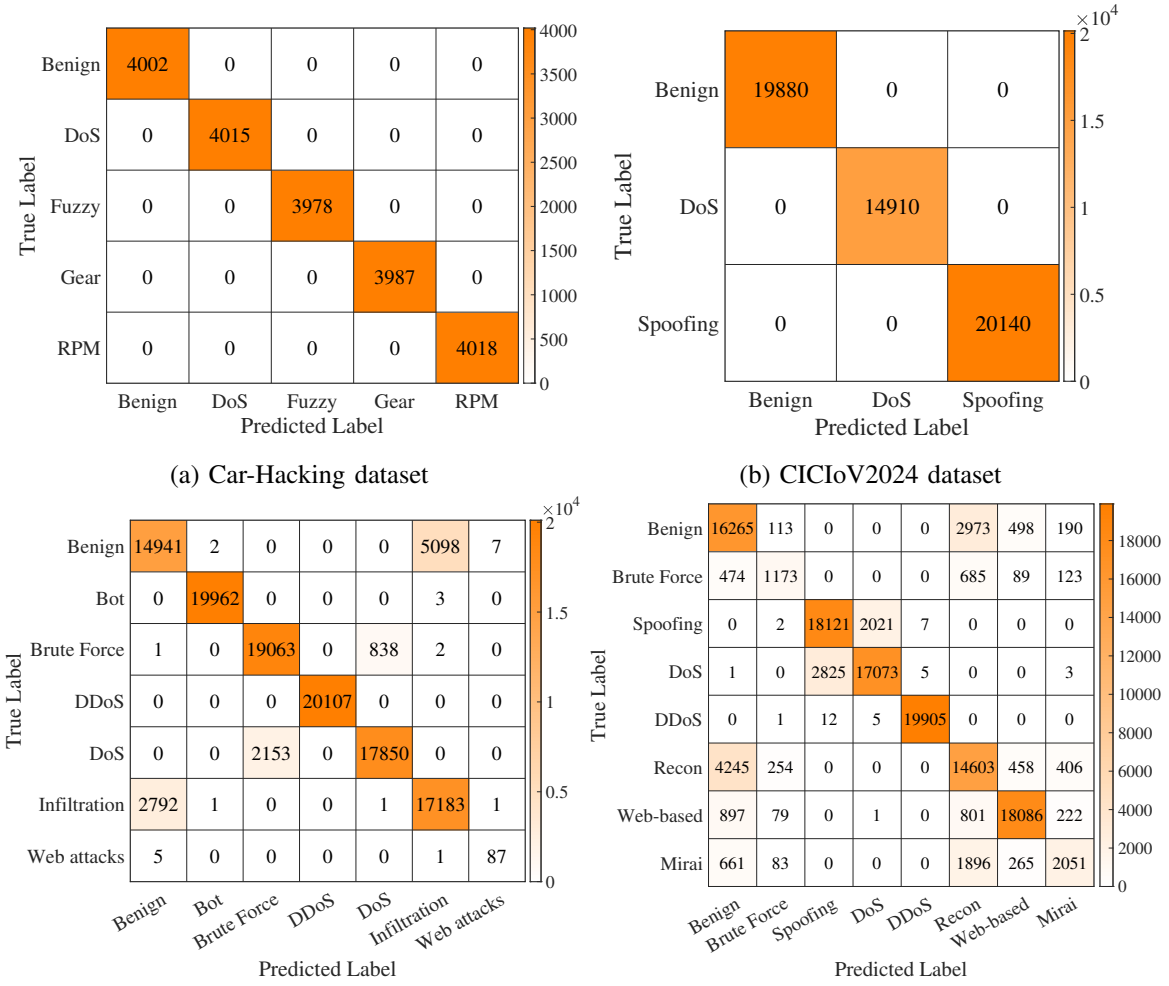


(a) Test accuracy on CICIoV2024 dataset



(b) Train loss on CICIoV2024 dataset

Fig. 8: Comparison of IVN performance indicators on CICIoV2024 dataset.



(a) Car-Hacking dataset

(b) CICIoV2024 dataset

(c) CICIDS2018 dataset

(d) CICIoT2023 dataset

Fig. 9: Confusion matrices of classification performance.

E. Cross-Network Intrusion Detection

To verify the effectiveness of our system in detecting cross-network attacks, we focused on DoS attacks in our simulation experiments due to their prevalence and severe impact on cross-network security. By simulating DoS attacks on the IVN datasets (Car-Hacking and CICIoV2024) and evaluating their effects on the EVN datasets (CICIDS2018 and CICIoT2023), we demonstrated the capability of CATwin-IDS in detecting cross-network threats. As shown in Table IX, CATwin-IDS achieved an F1 score of 0.9231 on the CICIDS2018 dataset and 0.8774 on the CICIoT2023 dataset, significantly outperforming other comparative systems. These results confirm that our solution effectively mitigates DoS attacks exploiting the IVN-EVN coupling, ensuring higher accuracy and reliability in EVN intrusion detection. Although our experiments primarily focused on DoS attacks, the design of CATwin-IDS positions it well to address other types of attacks. The system employs digital twin technology to construct dynamic models of both the IVN and EVN, enabling real-time monitoring and analysis of cross-network traffic anomalies. This digital twin-based detection mechanism not only enhances the detection of DoS attacks but also provides a robust foundation for identifying other attack types, such as data tampering and malicious injection. Specifically, digital twin technology can capture changes in network states in real-time, recognizing patterns of abnormal behavior rather than just specific attack signatures. Thus, even when confronted with unknown or emerging attack types, CATwin-IDS can promptly detect anomalies and respond accordingly. Additionally, the modular design of the

TABLE IX: F1 score comparison between CATwin-IDS and baseline models for cross-network DoS intrusion detection

Class Label	CICIoT2023 F1	CICIDS2018 F1
KNN	0.8539	0.8677
DT	0.8316	0.9200
ET	0.8446	0.9200
SVM	0.8506	0.9113
LightGBM	0.8686	0.8687
CatBoost	0.8683	0.9198
XGBoost	0.8692	0.9207
BERT	0.8658	0.9208
TinyBERT	0.8696	0.9203
CNN	0.8544	0.9126
LSTM	0.8612	0.9148
BiLSTM	0.8599	0.9142
RNN	0.8522	0.0530
CATwin-IDS	**0.8774**	**0.9231**

system allows for the easy integration of new detection algorithms or features as needed, further enhancing its adaptability and scalability. These design features indicate that CATwin-IDS, while excelling in DoS attack detection, also holds the potential to be extended to other attack types, offering comprehensive security protection for connected vehicles.

V. Conclusion

This paper addresses the challenge of cross-network intrusions arising from the tight coupling between IVN and EVN by proposing CATwin-IDS. The framework leverages DistilBERT with TSA for contextual feature modeling, employs CMI and Borderline-SMOTE for feature optimization and data balancing, and integrates digital twin technology to enable dynamic simulation and real-time synchronization. Experimental evaluations confirm that CATwin-IDS achieves higher detection performance and efficiency across multiple benchmark datasets, overcoming the limitations of existing IDS in handling cross-network attacks.

Future work will focus on enhancing the detection of complex cross-network attacks (e.g., Infiltration and Web-based threats) by enriching the diversity and fidelity of attack scenarios simulated in the digital twin environment, addressing the limited availability of rare attack samples. In addition, model compression and quantization techniques will be explored to further reduce inference latency and memory footprint for real-time deployment on in-vehicle edge devices. Finally, we plan to improve the interpretability of the detection results, for example through SHAP-based analysis, to provide deeper insights into feature contributions and model decision mechanisms.

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